

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication	20250268387
Kind Code	A1
Publication Date	August 28, 2025
Inventor(s)	D'Elia; Toby

Tanning Bed Foot Cushion

Abstract

A cushion device comprising a main body having an upper portion and a bottom portion; a plurality of protruding members attached to the upper portion; and an attachment member attached to the bottom portion.

Inventors:	D'Elia; Toby (San Rafael, CA)
Applicant:	D'Elia; Toby (San Rafael, CA)
Family ID:	1000007722795
Appl. No.:	18/587336
Filed:	February 26, 2024

Publication Classification

Int. Cl.:	A47G9/10 (20060101)
U.S. Cl.:	
CPC	A47C20/021 (20130101); A47G2009/1018 (20130101)

Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates generally to the field of comfort-enhancing accessories for indoor tanning facilities, and more specifically, to a novel cushioning device designed to improve the comfort and support of user foot positioning within indoor tanning beds.

BACKGROUND OF THE INVENTION

[0002] A cushioning device designed to improve the comfort and support of user foot positioning

within indoor tanning beds is in demand.

[0003] The utilization of indoor tanning beds is a popular method for individuals seeking to achieve a tanned appearance irrespective of outdoor weather conditions. While these devices offer convenience and controlled exposure to ultraviolet (UV) light, one common drawback experienced by users is the lack of comfort during tanning sessions.

[0004] Traditional tanning bed designs primarily focus on the functionality of UV light distribution. The surface on which the user lies is either flat or contoured, and is made of hard acrylic, neglecting the user's overall comfort and experience.

[0005] Existing solutions to enhance user comfort in tanning beds have been limited in scope. These solutions may include generic contoured and non-contoured head cushions that are not specifically designed for the ergonomic needs of the user's foot during tanning sessions.

[0006] As a result, there is a significant gap in the market for a product that addresses these specific needs, offering both enhanced comfort and support for the foot tailored to the indoor tanning context.

[0007] Additionally, the lack of customization in terms of aesthetics and functionality further limits the appeal and utility of such accessories within the competitive landscape of indoor tanning services. Thus, there is a need for an innovative solution that not only enhances user comfort, but incorporates aesthetic integration with existing tanning bed designs.

[0008] The present invention aims to overcome the limitations of traditional devices by introducing an innovative solution specifically engineered to deliver superior user comfort, durability, and aesthetic adaptability.

SUMMARY

[0009] In accordance with the present invention, a novel cushion device is provided. The cushion device comprises a main body having upper portion and a bottom portion; a first concavity and a second concavity attached to the upper portion, wherein the first concavity and the second concavity include an identical countered shape; and an attachment member attached to the bottom portion, wherein the attachment member includes at least one letter.

[0010] In one embodiment, the main body is made of a polyurethane foam.

[0011] In some embodiments, the attachment member includes letter forming "Tanshon."

[0012] In some embodiments, the main body includes an ultraviolet (UV) ray stabilizer additive.

[0013] In yet another embodiment, the main body includes a color additive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is an illustration of one embodiment of the present invention.

[0015] FIG. 2 is a top view of one embodiment of the present invention.

[0016] FIG. 3 is a side view of one embodiment of the present invention.

[0017] FIG. 4 is a side view of one embodiment of the present invention showing a first concavity and a second concavity, wherein each concavity is distinguished by varying heights.

[0018] FIG. 5 is a side view of one embodiment of the present invention showing troughs.

[0019] FIG. 6 is a bottom view of one embodiment of the present invention showing troughs.

[0020] FIG. 7 is a bottom view of one embodiment of the present invention showing troughs.

DETAIL DESCRIPTIONS OF THE INVENTION

[0021] All illustrations of the drawings are for the purpose of describing selected versions of the present invention and are not intended to limit the scope of the present invention.

[0022] As a preliminary matter, it will readily be understood by one having ordinary skill in the relevant art that the present disclosure has broad utility and application. As should be understood, any embodiment may incorporate only one or a plurality of the above-disclosed aspects of the

disclosure and may further incorporate only one or a plurality of the above-disclosed features.

[0023] Furthermore, any embodiment discussed and identified as being “preferred” is considered to be part of a best mode contemplated for carrying out the embodiments of the present disclosure. Other embodiments also may be discussed for additional illustrative purposes in providing a full and enabling disclosure. Moreover, many embodiments, such as adaptations, variations, modifications, and equivalent arrangements, will be implicitly disclosed by the embodiments described herein and fall within the scope of the present disclosure.

[0024] Accordingly, while embodiments are described herein in detail in relation to one or more embodiments, it is to be understood that this disclosure is illustrative and exemplary of the present disclosure, and are made merely for the purposes of providing a full and enabling disclosure. The detailed disclosure herein of one or more embodiments is not intended, nor is to be construed, to limit the scope of patent protection afforded in any claim of a patent issuing here from, which scope is to be defined by the claims and the equivalents thereof. It is not intended that the scope of patent protection be defined by reading into any claim limitation found herein and/or issuing here from that does not explicitly appear in the claim itself.

[0025] Additionally, it is important to note that each term used herein refers to that which an ordinary artisan would understand such term to mean based on the contextual use of such term herein. To the extent that the meaning of a term used herein—as understood by the ordinary artisan based on the contextual use of such term—differs in any way from any particular dictionary definition of such term, it is intended that the meaning of the term as understood by the ordinary artisan should prevail.

[0026] Furthermore, it is important to note that, as used herein, “a” and “an” each generally denotes “at least one,” but does not exclude a plurality unless the contextual use dictates otherwise. When used herein to join a list of items, “or” denotes “at least one of the items,” but does not exclude a plurality of items of the list. Finally, when used herein to join a list of items, “and” denotes “all of the items of the list.”

[0027] The following detailed description refers to the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the following description to refer to the same or similar elements. While many embodiments of the disclosure may be described, modifications, adaptations, and other implementations are possible. For example, substitutions, additions, or modifications may be made to the elements illustrated in the drawings, and the methods described herein may be modified by substituting, reordering, or adding stages to the disclosed methods.

[0028] Accordingly, the following detailed description does not limit the disclosure. Instead, the proper scope of the disclosure is defined by the claims found herein and/or issuing here from.

[0029] Through the integration of specialized materials and a meticulous manufacturing process, the present invention, Tanshon™, stands as a distinctive and valuable addition to the realm of indoor tanning accessories. The present invention's design and functionality address a previously unmet need for enhanced foot comfort and support for tanning bed users, marking a significant advancement in the user experience of indoor tanning facilities.

[0030] The present invention, referred to as the Tanshon, embodies a novel foam cushion specifically engineered for enhancing the comfort and support of user foot positioning within indoor tanning beds. This detailed description elucidates the structural components, material composition, and the manufacturing process of the Tanshon, delineating its innovative features and utility in the context of indoor tanning applications.

[0031] At its core, the Tanshon **100** comprises components, each selected for their specific properties and contributions to the functionality and durability of the final product.

[0032] The present invention provides a cushion device (Tanshon) **100** comprising a Polyurethane Foam, Ultraviolet Ray Stabilizer Additive, and Color Additive.

[0033] Serving as the primary material, polyurethane foam is chosen for its excellent cushioning

properties. This foam offers the necessary support and comfort for the feet, allowing users to adopt various resting positions without compromising on comfort during the tanning session.

[0034] The present invention has been meticulously developed with the capability to conform, deform, or become depressed in accordance with the pressures exerted by the user's feet, providing targeted support and thereby mitigating discomfort associated with prolonged periods of laying or resting.

[0035] In one embodiment, as shown in FIGS. 1-7, the cushion device **100** of the present invention may include a main body **10**. The main body **10** may include a base (bottom) portion **17** and an upper portion **15**, wherein the upper portion **15** is distinguished by a dual concavity design which may include a first concavity **20a** and a second concavity **20b**. The first and the second concavity **20a**, **20b** can be identical in shape.

[0036] In some embodiments, the first and the second concavity **20a**, **20b** can be different in shape. For example, as shown in FIG. 4, the first concavity **20a** may include a shallow height engineered to cradle the lower back or thighs, promoting spinal alignment or relieving pressure on the legs, respectively. The second concavity **20b** may include a deep height specifically tailored to accommodate the cervical region or knees, offering enhanced support and facilitating the alleviation of strain in these areas.

[0037] The present invention may include adaptive contouring capability, which enables the cushioning device **100** to distribute weight evenly and adapt to various foot shapes and sizes, thereby optimizing the user's comfort.

[0038] In some embodiments, the present invention may also feature a plurality of anti-slip elements strategically positioned on the base portion's underside, ensuring the device remains stationary during use, thereby enhancing safety and usability.

[0039] In preferred embodiment, the main body **10** is made of a polyurethane foam and includes a plurality of protruding members **20a**, **20b** and a bottom attachment **30**.

[0040] In one embodiment, the main body **10** can be an elongated panel having two protruding members **20a**, **20b** attached to each end of the elongated panel, main body **10**.

[0041] In some embodiments, the two protruding members **20a**, **20b** can include a contoured shape.

[0042] The bottom attachment **30** can be attached to the bottom surface (bottom portion) **17** of the elongated panel **10**. The bottom attachment **30** can include a plurality of letters.

[0043] In some embodiments, as shown in FIG. 7 the bottom attachment **30** may feature characters that spell out the word "Tanshon." In such embodiments, the alphanumeric indicia (letters) may be secured in such a fashion that they project, with a predetermined height, from the bottom surface of the cushion device **100**. In alternative embodiments, the letter may be configured in a manner whereby they are inset or indented into the bottom surface of the cushion device **100**, thereby offering a distinct structural configuration.

[0044] In certain embodiments of the present invention, as shown in FIG. 5 and FIG. 6, the cushion device **100** may further comprise concave formations, hereinafter referred to as troughs **40**, strategically positioned between the alphanumeric indicia (letters) of the bottom attachment **30** positioned on the lower segment (bottom portion) **17** of cushion device **100**. These troughs **40** can be specifically engineered to enhance the pliability and flexibility of the cushion device **100**. The troughs **40** can be extending across the full width of the body member **10** or may be elongated to any length deemed appropriate. Furthermore, the number of the troughs **40** can be adjusted to adequately accommodate a variety of flat surfaces to which the body member **10** is to be attached.

[0045] In some embodiments, the troughs **40** can include a rounded configuration extending amidst the alphanumeric indicia **30**, thereby serving as zones for alleviating mechanical stress. This is achieved through the reduction of surface tension on the lower segment (bottom portion) **17** of the cushion device **100**, which, in turn, diminishes the magnitude of force required to induce bending and flexing of the cushion device **100** under load. Consequently, this innovative design mitigates the necessity for the bottom portion **17** to undergo extensive stretching in the aforementioned areas.

[0046] In some embodiments, this feature is instrumental in further optimizing the mechanical performance and durability of the cushion device **100** by facilitating a more nuanced distribution of stress and enhancing the overall structural integrity of the cushion device **100**.

[0047] In preferred embodiment, to ensure the longevity and maintenance of the present invention, an ultraviolet (UV) ray stabilizer additive is incorporated into the material composition. This additive plays a crucial role in deflecting UV rays, which are inherently abundant in tanning bed environments, thereby preventing potential harm or degradation to the product over time.

[0048] In preferred embodiment, to cater to aesthetic preferences and to facilitate seamless integration with existing tanning bed designs, the present invention incorporates a color additive. This additive allows for the production of the cushion device **100** in two standard color options: grey and black.

[0049] In some embodiments, the present invention can be made by the following method which can be a systematic and precise manufacturing process.

[0050] A product Computer-Aided Design (CAD) file is provided to the manufacturer, detailing the specific dimensions and contours of the cushion device.

[0051] Utilizing the CAD file, the manufacturer proceeds to create a mold that accurately reflects the design specifications of the cushion device **100**.

[0052] Concurrently, the procurement of raw materials—polyurethane foam, UV ray stabilizer additive, and color additive—is arranged, ensuring all components meet predefined quality standards.

[0053] The raw materials are then scaled and portioned according to the desired formula, tailored to achieve the optimal composition for cushioning, durability, and color consistency.

[0054] Following the preparation of materials, they are combined and thoroughly mixed to ensure a homogeneous mixture.

[0055] The resultant liquid mixture is then carefully poured into the prepared mold, initiating the molding process.

[0056] The mixture is allowed to set within the mold for a duration of 5-10 minutes, a critical period during which the foam begins to solidify and take shape.

[0057] Upon completion of the setting time, the mold is opened, and the newly formed cushion device **100** is carefully removed.

[0058] The final step involves the manual trimming of any excess material along the edges and a comprehensive quality check to ensure the product meets all specified criteria.

[0059] Although the invention has been explained in relation to its preferred embodiment, it is to be understood that many other possible modifications and variations can be made without departing from the spirit and scope of the invention.

Claims

1. A cushion device comprising: a main body having upper portion and a bottom portion, the main body is an elongated panel having a length longer than a width, the length is sized to allow a person to lay back on the main body; a plurality of protruding members, having a convex shape, attached to the upper portion, wherein the distance between any part of any two protruding members is greater than the length of the main body, wherein the plurality of protruding members have an elongated length extending across the width of the main body, sized to support a person laying on the plurality of protruding members; and a bottom attachment attached to the bottom portion.
2. The cushion device as claimed in claim 1, wherein the plurality of protruding members include a contoured shape.
3. The cushion device as claimed in claim 2, wherein the main body is made of a polyurethane foam.
4. The cushion device as claimed in claim 3, wherein the bottom attachment includes letters

forming a word “Tanshon.”

5. The cushion device as claimed in claim 4, wherein the letters are indented into the bottom portion.

6. A cushion device comprising: a main body having upper portion and a bottom portion, the main body is an elongated panel having a length longer than a width, the length is sized to allow a person to lay back on the main body; a first concavity and a second concavity attached to the upper portion, wherein the distance between any part of the first concavity and the second concavity is greater than the length of the main body, wherein the first concavity and the second concavity have an elongated length extending across the width of the main body, sized to support a person laying on the first concavity and the second concavity; and a bottom attachment attached to the bottom portion, wherein the bottom attachment includes a word ‘Tanshon,’ forming troughs between the letters in the word.

7. The cushion device as claimed in claim 6, wherein the first concavity and a second concavity include a contoured shape.

8. The cushion device as claimed in claim 7, wherein the first concavity and a second concavity are identical.

9. The cushion device as claimed in claim 8, wherein the main body is made of a polyurethane foam.

10. cancelled.

11. cancelled.

12. cancelled.

13. A cushion device comprising: a main body having upper portion and a bottom portion, the main body is an elongated panel having a length longer than a width, the length is sized to allow a person to lay back on the main body; a first concavity and a second concavity attached to the upper portion, wherein the first concavity and the second concavity include an identical contoured shape, wherein the distance between any part of the first concavity and the second concavity is greater than the length of the main body, wherein the first concavity and the second concavity have an elongated length extending across the width of the main body, sized to support a person laying on the first concavity and the second concavity; a bottom attachment attached to the bottom portion, wherein the bottom attachment includes a plurality of letters; and a plurality of troughs placed in the bottom portion of the main body between the letters.

14. The cushion device as claimed in claim 13, wherein the main body is made of a polyurethane foam.

15. The cushion device as claimed in claim 14, wherein the bottom attachment includes letter forming “Tanshon.”

16. The cushion device as claimed in claim 15, wherein the main body includes an ultraviolet (UV) ray stabilizer additive.

17. The cushion device as claimed in claim 13, wherein the main body includes a color additive.

18. The cushion device as claimed in claim 17, wherein the color additive includes a black color.

19. The cushion device as claimed in claim 17, wherein the color additive includes a grey color.
